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ECEN 474/704
Lab 9: Prelab
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1. **Two-stage Miller operational-amplifier design (Fig. 9-3).** Design the amplifier to meet Table 9-1 of the laboratory handout.

Parameter	Requirement
Open-loop DC voltage gain, A_{v0}	$> 55\,\text{dB}$
Common-mode rejection ratio	$> 60\,\text{dB}$
Gain–bandwidth product	$> 3\,\text{MHz}$
Phase margin in unity-gain feedback	$> 45^\circ$
Peak-to-peak output swing	$> 1.2\,\text{V}$
Load capacitance, C_L	$30\,\text{pF}$
Load resistance, R_L	∞ (open circuit)
Power, including the bias-reference branch	$< 500\,\mu\text{W}$
Power supplies	$V_{DD} = +0.9\,\text{V}, \quad V_{SS} = -0.9\,\text{V}$

Solution.

Device parameters and assumptions. The device parameters are taken from the single-finger $W_{\text{total}} = 5\,\mu\text{m}$, $L = 0.27\,\mu\text{m}$ measurements in Table I of the Lab 1 report:

Parameter	NFET	PFET
$K_P = \mu C_{ox}$ ($\mu\text{A/V}^2$)	215.6	47.57
$\lvert V_{TH}\rvert$ (V)	0.4330	0.4068
λ (V^{-1})	0.1624	0.1945
θ (V^{-1})	0.9999	0.6558

All transistors have $\boxed{L = 0.270\,\mu\text{m}}$. The square-law model gives

$$I_D \approx \frac{1}{2} K_P \frac{W_{\text{eff}}}{L} V_{OV}^2, \qquad g_m \approx \frac{2I_D}{V_{OV}}, \qquad r_o \approx \frac{1}{\lambda I_D}.$$

The DC analysis neglects body effect, mobility degradation, and channel-length modulation. Thus, the listed θ does not enter the sizing equations. Finite r_o is retained in the small-signal model, while intrinsic node capacitances are neglected in the initial frequency analysis. The results below follow from these approximations.

(a) Current budget and transconductance selection.

In Fig. 9-3, M1/M2 form the NMOS input pair, M3/M4 the PMOS mirror load, M5 the diode-connected NMOS bias reference, M6 the tail source, M7 the output sink, and M8 the PMOS gain device. C_c connects the first-stage output to the final output.

Including the reference branch, the power limit imposes

$$I_{\text{total}} < \frac{500\,\mu\text{W}}{V_{DD} - V_{SS}} = \frac{500\,\mu\text{W}}{1.8\,\text{V}} = 277.778\,\mu\text{A}.$$

The selected bias currents and compensation capacitance are

$$\boxed{I_{\text{bias}} = 10\,\mu\text{A}, \quad I_T = 40\,\mu\text{A}, \quad I_2 = 160\,\mu\text{A}, \quad C_c = 10\,\text{pF}.}$$

At balanced input, $I_{D1,2,3,4} = 20\,\mu\text{A}$. The ideal current ratios are M6:M5 = 4 : 1 and M7:M5 = 16 : 1, giving

$$P_Q = (1.8\,\text{V})(10 + 40 + 160)\,\mu\text{A} = \boxed{378\,\mu\text{W} < 500\,\mu\text{W}}.$$

The nominal power margin is therefore $122\,\mu\text{W}$. A DC common-mode sweep determines the maximum power over the valid input range.

From GBW $\approx g_{m1}/(2\pi C_c)$,

$$g_{m1} > 2\pi(3\,\text{MHz})(10\,\text{pF}) = 188.50\,\mu\text{S}.$$

An input overdrive of $V_{OV1,2} = 0.160\,\text{V}$ gives $g_{m1,2} = 250\,\mu\text{S}$. For the remaining devices, the selected overdrives are $V_{OV3,4,8} = 0.200\,\text{V}$ and $V_{OV5,6,7} = 0.150\,\text{V}$, yielding

$$g_{m3,4} = 200\,\mu\text{S}, \qquad g_{m8} = 1.600\,\text{mS}.$$

The resulting estimate is $g_{m1}/(2\pi C_c) = 3.979\,\text{MHz}$. Part (f) includes the second pole and the feedforward zero in the unity-gain frequency and phase-margin calculation.

(b) Continuous analytical dimensions. Solving the drain-current equation for the aspect ratio, with currents in μA and K_P in $\mu\text{A/V}^2$, gives

$$\frac{W_{\text{eff}}}{L} = \frac{2I_D}{K_P V_{OV}^2}.$$

For example, $(W/L)_{1,2} = 40/[215.6(0.16)^2] = 7.24722$. The resulting continuous widths are:

Devices	Type	I_D (μA)	V_{OV} (V)	W_{eff} (μm)	L (μm)
M1, M2	NFET	20	0.160	1.95675	0.270
M3, M4	PFET	20	0.200	5.67585	0.270
M5	NFET	10	0.150	1.11317	0.270
M6	NFET	40	0.150	4.45269	0.270
M7	NFET	160	0.150	17.81076	0.270
M8	PFET	160	0.200	45.40677	0.270

(c) Grid-constrained initial sizes and dummy arrangement.

The analytical dimensions are rounded to the 0.01 μm design grid. For a finger width w, a finger count n_f, and a parallel multiplicity m, the effective width is

$$W_{\text{eff}} = w\, n_f\, m, \qquad w_t = w\, n_f.$$

The instance width w_t excludes m. The entries below apply to each named transistor individually.

Device	w (μm)	L (μm)	n_f	m	W_{eff} (μm)
M1, M2	0.98	0.27	1	2	1.96
M3, M4	1.42	0.27	1	4	5.68
M5	1.11	0.27	1	1	1.11
M6	1.11	0.27	1	4	4.44
M7	1.11	0.27	1	16	17.76
M8	1.42	0.27	1	32	45.44

These dimensions preserve the NMOS bias ratios exactly and give $(W/L)_8/(W/L)_3 = 8$. Hence, the 160 μA output PMOS and 20 μA mirror devices have equal square-law overdrive. The gate of M8 is driven by the first-stage output and is separate from the diode-connected gate of M3.

For $I_{\text{bias}} = 10\,\mu$A, the rounded dimensions give

$$V_{OV1,2} = 0.159867\,\text{V}, \quad V_{OV3,4,8} = 0.199927\,\text{V}, \quad V_{OV5,6,7} = 0.150214\,\text{V}.$$

Thus $g_{m1,2} = 250.208\,\mu$S and $g_{m8} = 1.600585\,$mS. The subsequent calculations use these dimensions.

Matching and dummies. M1/M2 use equal units in an AB/BA common-centroid arrangement. M3/M4 likewise use equal unit widths, equal centroids, and symmetric routing. M5/M6/M7 use the same unit size and orientation; their different multiplicities set the intended current ratios. The resulting arrangement contains 65 active one-finger units.

Each active MOS symbol has one adjacent dummy *symbol* with the same unit w and L. A dummy multiplicity of $n_f = 1$, $m = 2$ represents two physical edge units per active device group (16 dummy units in total). NFET dummy D/G/S/B all connect to V_{SS}; PFET dummy D/G/S/B all connect to V_{DD}. The physical edge units follow the boundaries of the matched groups. Since the dummies carry no signal current, their widths are excluded from active W_{eff}.

Consistent schematic and extracted netlists require the same finger widths, finger counts, and multiplicities. This includes the CDF width convention and the extraction `mSwitch` setting. The dimensions above define the initial design; pre-layout DC and AC simulations determine the final dimensions.

(d) Operating point, common-mode range, and output headroom.

The selected input bias is $V_{CM,Q} = 0.150\,\text{V}$, with a nominal output target of $V_{o,Q} = 0$. Let v_x be the M1/M3 drain node, v_y the M2/M4 drain and M8 gate node, and v_s the input-pair source node. Substitution of the rounded dimensions gives

$$\begin{aligned} V_B &= V_{SS} + V_{TN} + V_{OV5} = -0.316786\,\text{V}, \\ V_s &= V_{CM,Q} - V_{TN} - V_{OV1} = -0.442867\,\text{V}, \\ V_x &= V_{DD} - |V_{TP}| - V_{OV3} = 0.293273\,\text{V}, \\ V_y &= V_{DD} - |V_{TP}| - V_{OV8} = 0.293273\,\text{V}. \end{aligned}$$

At this output voltage, $V_{DS7} = V_{SD8} = 0.9\,\text{V}$. Also, $V_{DS1,2} = 0.736140\,\text{V}$, $V_{SD3,4} = 0.606727\,\text{V}$, and $V_{DS6} = 0.457133\,\text{V}$, all above their selected overdrives. With channel-length modulation neglected, equal output-branch currents do not uniquely determine $V_{o,Q}$. An input-offset sweep establishes the bias that produces $V_o = 0$ for open-loop measurements.

The input-pair and tail-source saturation conditions give

$$\begin{aligned} V_{CM,\text{min}} &\approx V_{SS} + V_{OV6} + V_{TN} + V_{OV1} = -0.156919\,\text{V}, \\ V_{CM,\text{max}} &\approx V_{DD} - |V_{TP}| - V_{OV3} + V_{TN} = 0.726273\,\text{V}. \end{aligned}$$

The selected 0.150 V lies inside this interval. These bounds follow from the stated threshold-voltage approximation; the transistor-level DC sweep includes body effect and the associated bias shifts.

At the output, the saturation-limited window is

$$V_{SS} + V_{OV7} < V_o < V_{DD} - V_{OV8}, \qquad -0.749786 < V_o < 0.700073\,\text{V}.$$

Its total span is 1.449859 V, and the symmetric headroom about zero is approximately $\boxed{1.400146\,\text{V}_{pp} > 1.2\,\text{V}_{pp}}$. This interval is set by transistor saturation. The swing at a specified distortion is obtained from 100 Hz open-loop PSS at the input amplitude corresponding to 5% THD.

Slew rate. The tail current limits the rate of change of the compensation capacitor voltage, giving

$$\text{SR} \approx I_T/C_c = \boxed{4.0\,\text{V}/\mu\text{s}}.$$

For falling output, M7 must sink both compensation and load current:

$$\frac{I_2 - I_T}{C_L} = \frac{160 - 40}{30}\frac{\mu\text{A}}{\text{pF}} = 4.0\,\text{V}/\mu\text{s}.$$

A rising $4\,\text{V}/\mu\text{s}$ output requires about $I_8 = I_7 + I_T + C_L\text{SR} = 320\,\mu\text{A}$, twice its quiescent current. The rising slew-rate estimate therefore requires M8 to supply this current. Transient analysis determines the rising and falling slew rates; Lab 9 specifies no numerical minimum.

(e) Finite-output-resistance gain and CMRR.

The common-mode response depends on the finite tail-source conductance. For the small-signal analysis, define

$$n = g_{m1},\ p = g_{m3},\ g = g_{ds1},\ q = g_{ds3},\ t = g_{ds6},\quad u = g + q,\quad h = n + g.$$

Matched pairs have equal small-signal parameters. With the supplies and bias gate at AC ground, the first-stage nodal matrix is

$$\begin{bmatrix} u+p & 0 & -h \\ p & u & -h \\ -g & -g & t+2h \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_s \end{bmatrix} = \begin{cases} [n/2,\ -n/2,\ 0]^T v_{id}, \\ [-n,\ -n,\ 2n]^T v_{cm}. \end{cases}$$

Here $v_{id} = v_{i+} - v_{i-}$, with M2 receiving $+v_{id}/2$. Let $D = t(u+p) + 2h(p+q)$. Solving gives

$$A_{d1} = \frac{v_y}{v_{id}} = -\frac{n[t(u+2p) + 2h(2p+q)]}{2uD},$$
$$A_{cm1} = \frac{v_y}{v_{cm}} = -\frac{nt}{D}.$$

For the grid sizes, in μS,

$$n = 250.2076,\quad p = 200.0732,\quad g = 3.248,\quad q = 3.890,\quad t = 6.496.$$

Therefore

$$A_{d1} = -34.71517,\qquad A_{cm1} = -0.01551834,$$
$$\boxed{\mathrm{CMRR} = 20\log_{10}\left|\frac{A_{d1}}{A_{cm1}}\right| = 66.9935\,\mathrm{dB} > 60\,\mathrm{dB}.}$$

The second-stage gain cancels in this ratio. The result retains finite tail r_o under the assumptions of matched devices and negligible body effect.

The first-stage output resistance is found by setting both input voltages to zero and injecting a test current into v_y:

$$R_A = \frac{(u+p)(t+2h) - gh}{uD} = 141.1964\,\mathrm{k\Omega}.$$

Thus its equivalent differential transconductance is $G_{mA} = |A_{d1}|/R_A = 245.8644\,\mu\mathrm{S}$. For the second stage,

$$R_B = r_{o7} \parallel r_{o8} = \frac{1}{(\lambda_n + \lambda_p) I_2} = 17.5119\,\mathrm{k\Omega},$$
$$A_2 = -g_{m8} R_B = -28.02930,$$
$$A_{v0} = A_{d1} A_2 = 973.0421\,\mathrm{V/V},$$
$$20\log_{10}|A_{v0}| = \boxed{59.7626\,\mathrm{dB} > 55\,\mathrm{dB}}.$$

(f) Miller compensation and phase margin.

The first stage is represented by its Norton equivalent from part (e), with $C_A = 0$, $C_B = C_L = 30\,\text{pF}$, and $C_c = 10\,\text{pF}$. The two-node model gives

$$H(s) = \frac{A_{v0}(1 - s/\omega_z)}{1 + as + bs^2}, \qquad \omega_z = \frac{g_{m8}}{C_c},$$

$$a = R_A(C_A + C_c) + R_B(C_B + C_c) + g_{m8}R_AR_BC_c,$$
$$b = R_AR_B(C_AC_B + C_AC_c + C_BC_c).$$

For the selected dimensions,

$$a = 4.1688808 \times 10^{-5}\,\text{s}, \qquad b = 7.4178551 \times 10^{-13}\,\text{s}^2.$$

The right-half-plane zero adds phase lag. The two poles are $s = -\omega_{p1}$ and $s = -\omega_{p2}$:

$$\omega_{p1} = \frac{2}{a + \sqrt{a^2 - 4b}}, \qquad \omega_{p2} = \frac{a + \sqrt{a^2 - 4b}}{2b}.$$

Converting angular frequencies to Hz,

$$\boxed{f_{p1} = 3.81932\,\text{kHz}, \quad f_{p2} = 8.94079\,\text{MHz}, \quad f_z = 25.47411\,\text{MHz}.}$$

Gain–bandwidth product and unity-gain frequency. The bandwidth f_{BW} satisfies $|H(j2\pi f_{\text{BW}})| = A_{v0}/\sqrt{2}$, and the gain–bandwidth product is $\text{GBW} = A_{v0}f_{\text{BW}}$. The unity-gain frequency f_u satisfies $|H(j2\pi f_u)| = 1$. With $x = \omega_u^2$, this condition reduces to

$$b^2x^2 + \left(a^2 - 2b - \frac{A_{v0}^2}{\omega_z^2}\right)x + 1 - A_{v0}^2 = 0.$$

Taking its positive solution yields

$$f_{\text{BW}} = 3.81932\,\text{kHz},$$
$$\text{GBW} = \boxed{3.71636\,\text{MHz} > 3\,\text{MHz}},$$
$$f_u = \boxed{3.49386\,\text{MHz}}.$$

For unity negative feedback,

$$\text{PM} = 180^\circ - \tan^{-1}\frac{f_u}{f_{p1}} - \tan^{-1}\frac{f_u}{f_{p2}} - \tan^{-1}\frac{f_u}{f_z} = \boxed{60.9086^\circ > 45^\circ}.$$

The two-node model therefore satisfies the phase-margin requirement. The calculation excludes intrinsic mirror and tail-node poles, body effect, and layout parasitics. Unity-feedback STB analysis evaluates their effect on the complete transistor circuit.

(g) Summary.

All values below use the grid-constrained geometry in part (c).

Quantity	Calculated value	Specification / note
DC gain	59.76 dB	> 55 dB
CMRR	66.99 dB	> 60 dB
GBW	3.716 MHz	> 3 MHz
Unity-gain frequency	3.494 MHz	Actual model crossing
Phase margin	60.91°	$> 45°$
Power including bias	378 μW	$< 500\,\mu$W
Centered output headroom	1.400 V_{pp}	$> 1.2\,V_{pp}$ target
5%-THD PSS swing	To be simulated	Not established by headroom
Common-mode input range	$[-0.157,\ 0.726]$ V	Body effect omitted
Slew-rate estimate	4.0 V/μs	No specified minimum
C_c, C_L, R_L	10 pF, 30 pF, ∞	Selected compensation/load

With $C_A = 1$ pF at the first-stage output, the two-node model gives GBW = 3.704 MHz, f_u = 3.429 MHz, and PM = 58.97°. Thus, the selected compensation maintains both the gain–bandwidth and phase-margin requirements with this additional node capacitance.

Within the model used here, the calculated gain, CMRR, GBW, phase margin, and quiescent power satisfy the specified limits. The dimensions in part (c), together with $C_c = 10$ pF, therefore form the initial design. The 1.400 V_{pp} output headroom exceeds the 1.2 V_{pp} swing target. This headroom follows from the saturation constraints; the output swing at 5% THD is determined by the 100 Hz PSS test.

The closed-loop IM3 test uses 10 kHz and 11 kHz tones with an amplitude of 300 mV each. The sum of the tone amplitudes is 0.6 V, greater than the available input excursion about V_{CM} = 0.150 V. The specified test therefore extends beyond the calculated common-mode input range. PSS determines the resulting intermodulation products, while noise, PSRR, and sensitivity to process variation and mismatch are evaluated in their respective transistor-level analyses.

References.

[1] *ECEN 474/704 Lab 9: Two-Stage Miller Operational Amplifier*, course handout: Fig. 9-3, small-signal model, and Table 9-1; PSS swing and report instructions.

[2] Anonymous contributor, *ECEN 474/704 Lab 1 Report*, Table I: single-finger $W = 5\,\mu$m, $L = 0.27\,\mu$m NFET/PFET parameter rows.